
ECE228 Project Proposal: Physics-informed Deep Learning in Fluid Dynamics

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1 Problem Background & Motivation

Generally speaking, modeling complex non-linear system is either intractable on paper or very computationally expensive. Among all the non-linear physical problems, fluid dynamics often involves extensive use of computer modeling due to its complex and chaotic behavior, which is governed by Navier–Stokes equations. Navier–Stokes equations is a cornerstone of engineering design, aerodynamics, biomedical devices, and industrial systems. Traditional numerical solvers such as finite volume methods (e.g., ANSYS Fluent) are highly accurate but computationally expensive, requiring mesh discretizations and iterative solvers that scale poorly with problem complexity. Pure neural network could fit the experimental data easily, but can results in physically impossible solutions and usually requires massive labeled datasets which is often unavailable.

2 Related Work

Many methods have been proposed to address this common physical problems over the years:

- **Convolution Neural Network** Earlier days, convolutional Networks are proposed with a highly tailored architecture, trained using a novel unsupervised learning framework to solve the linear system (1).
- **Data-driven Projection Method** Traditional data-driven methods have also been improved on. For example, novel data-driven projection method using an artificial neural network to avoid iterative computation. Once the grid resolution is decided, data-driven method could obtain projection results in relatively constant time per grid cell, which is independent of scene complexity (2).
- **Encoder-decoder Network** Recently, some works propose convolutional encoder-decoder neural network as well as a conditional flow-based generative model for the solution of PDEs. Instead of residual function, KL divergence is between the model predictive density and the reference conditional density were used as loss function to optimize the model (3).

3 High-level Methodology

Physics-Informed Neural Networks (PINNs) (4) have emerged as a mesh-free alternative that embeds physical laws directly into the traditional neural network training process via specially designed loss function. PINN eliminates the need for labeled simulation data and constrains the network’s trainable parameters within a feasible solution space.

This is done by sampling a set of input spatiotemporal variables ($\{(x_j, y_j)\}$), passing them through the network and outputs stream function ψ , Cauchy stress tensor σ and pressure p . Next, gradients of

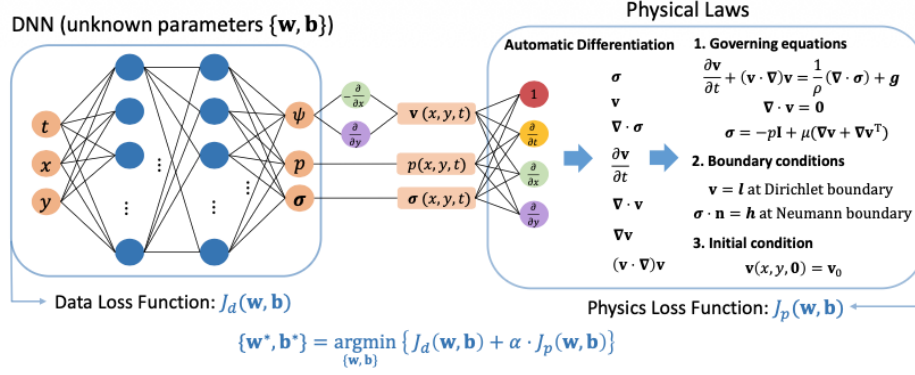


Figure 1: Predictions for target station 1 and 2

the network's output with respect to its input are computed at these locations. Then, the automatic differentiation is used to obtain the partial derivatives of the DNN output regarding the time and space.

For optimization, the loss function is composed of the data loss J_d and the physics loss J_p - where the J_d refers to the loss of DNN with unknown weights and J_p indicates the constraint of initial and boundary conditions converted as residuals which is added to the loss function based on Lagrangian multiplier. Physics loss is defined as $J_p = J_g + \beta J_{i/bc}$.

Improvement

We can replace the fixed boundary condition weight α and β with an adaptive loss weighting scheme, where they are updated at each iteration based on the gradient statistics of each loss term, eliminating the need for manual tuning and improving robustness across problem settings.

References

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